

COS : A-Series — Decision Architecture

senkaL8

Chapter 1: Why Decision-Making Fails

We make decisions every day.

What to choose, how to act, and what to prioritize are not determined solely by ability or knowledge.

Consider a statement such as, “I want to travel alone.”

In many cases, this would be interpreted as a request for travel-related advice. Many LLMs and human advisors could immediately proceed to planning accommodations, transportation, budget, and itinerary.

However, the meaning of such support changes completely when the underlying assumptions differ. The speaker may be a minor. They may have an illness or physical limitations that make independent travel difficult. They may also be facing circumstances that require more urgent attention than travel.

The problem is not that the proposed plan is necessarily wrong. The problem is that decision-making has begun before the relevant assumptions have been confirmed.

Even when the statement itself remains the same, different assumptions lead to different options, priorities, and permissible actions.

The same phenomenon can be observed in situations involving conflict. For example, suppose someone says, “My computer data was lost. I demand compensation and damages.” Such a statement is easily interpreted as a financial demand. In reality, however, what the person may want most is not money, but the restoration of the lost data.

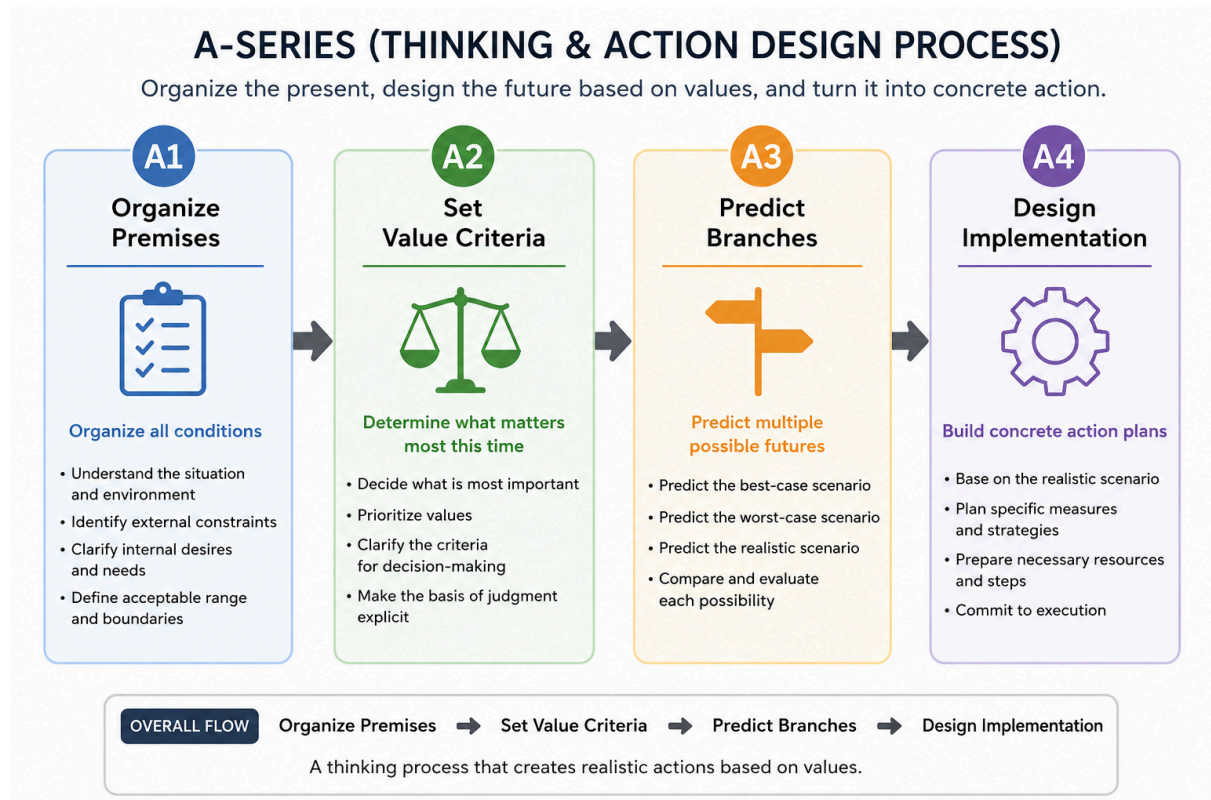
This issue is not limited to human-to-human communication. It may also be present in many contemporary dialogue systems, including current LLMs.

Furthermore, assumption mismatches are not merely communication errors. When decisions are made before assumptions are aligned, subsequent choices, priorities, and action plans may all be built upon an incorrect foundation.

What matters in decision-making is not only the decision itself, but also the assumptions upon which that decision is based.

The A-Series is a model that organizes this process into four stages of decision-making.

Chapter 2: What Is the A-Series?



The A-Series is the thought and action design module within COS (Cognitive Operating System).

Its role is not to produce correct conclusions. It is to organize the process by which decisions are formed and to decompose that process into observable components.

Decision-making is often perceived as a single act. In reality, however, multiple distinct processes exist before a judgment is reached.

The A-Series organizes this process into four functions.

- A1: Premise Clarification
- A2: Value Criteria
- A3: Divergence Prediction
- A4: Implementation Design

A1 clarifies the premises that form the foundation of judgment. A2 makes explicit the value criteria that determine what is prioritized. A3 predicts the multiple possible outcomes that arise from a given choice. A4 converts the selected direction into an actionable form.

These functions do not necessarily execute in a single linear sequence. In actual decision-making, they may interact and cycle back on one another.

However, by decomposing the process into these four functions, it becomes easier to identify where judgment became distorted and where room for improvement exists.

The purpose of the A-Series is not to guarantee correct conclusions. It is to make the decision-making process observable as a structure.

Chapter 3: A1 — Premise Clarification

A1 is the function of clarifying "what is treated as premise" before judgment begins.

A1 is placed first because, without clear premises, the value criteria and divergence predictions that follow cannot be stable. No matter how refined the judgment, if the foundation is unstable, deviation begins at the starting point.

In A1, premises such as situational and environmental conditions, external constraints, internal needs, and tolerance boundaries are organized to make the starting point of judgment observable. These are representative observation items referenced when confirming premises — they do not constitute a strict classification system.

The purpose of A1 is not to derive correct premises. It is to make explicit what has been placed as the foundation of judgment.

Chapter 4: A2 — Value Criteria

Chapter 4: A2 – Value Criteria

A2 is the function that clarifies the evaluation criteria used to determine what should be prioritized based on the assumptions organized in A1.

Even for the same person, the evaluation criteria that receive priority may change when the underlying assumptions change. For example, suppose a person has the following conditions identified in A1: it is the weekend, they have no scheduled plans, they want a change of pace, they want to drink coffee, and an exam is approaching. If the weather is clear, they may prioritize recreation or getting coffee and decide to go out. However, if it begins to rain, they may instead prioritize comfort or avoiding inconvenience, or they may cancel the outing altogether and prioritize studying for the exam. In this way, changes in assumptions can alter which value criteria are given priority.

A2 does not deal with values themselves. Rather, it deals with the evaluation criteria used to determine what should be prioritized in a decision-making process. A2 makes those priorities explicit under the current set of assumptions.

A2 does not exist independently of A1. The value criteria handled by A2 are not intended to classify personality traits. Instead, they may change through interaction with the assumptions organized in A1.

The purpose of A2 is not to analyze values. Its purpose is to make observable what is being prioritized under the assumptions identified in A1.

Chapter 5: A3 — Divergence Prediction

A3 is the function that predicts multiple possible outcomes based on the value criteria clarified in A2.

In decision-making, people often proceed by focusing on a particular expected outcome. A3, however, does not converge on a single scenario. Instead, it maintains multiple possible futures in parallel through three perspectives: realistic, best-case, and worst-case outcomes.

The purpose is not to prepare for the worst-case scenario. Rather, by maintaining multiple possible outcomes simultaneously, A3 prevents the action design process in A4 from becoming a bet on a single scenario.

The three perspectives can be defined as follows. A realistic outcome is the outcome that appears most likely based on the information currently available. A best-case outcome is the outcome in which the value criteria defined in A2 are satisfied to the greatest extent. A worst-case outcome is the outcome in which those value criteria are not only unmet but are driven in the opposite direction. These three outcomes are not fixed. If the priorities defined in A2 change, the evaluation of what constitutes the best-case or worst-case outcome may also change.

Consider the situation of being invited to a party. Suppose A2 has clarified the value criterion of prioritizing an enjoyable experience. The realistic outcome is the situation that appears most likely based on the participants, the atmosphere of the event, and past experiences. The best-case outcome is spending time with compatible people and having a highly satisfying experience. The worst-case outcome is not enjoyment at all, but interpersonal conflict, exhaustion, or damage to relationships.

A3 does not select one of these outcomes as the correct answer. Instead, it maintains multiple possibilities simultaneously so that A4 can design actions that remain viable across multiple situations.

The purpose of A3 is not to predict the correct future. Its purpose is to create a state in which action design in A4 can respond to multiple possible outcomes.

Chapter 6: A4 — Implementation Design

A4 is the function that designs executable actions based on the multiple outcomes maintained in A3.

The role of A4 is not to choose a single future and bet on it. Rather, it designs actions that aim for the best-case outcome, avoid the worst-case outcome, and remain capable of responding even if the worst-case outcome occurs.

Consider the party example. Suppose A2 has clarified the value criterion of prioritizing an enjoyable experience, and A3 has maintained realistic, best-case, and worst-case outcomes.

One action that may be designed in A4 is the question, “Who will be there?” This is not a simple confirmation. It is an example of an action designed to help realize the best-case outcome and avoid the worst-case outcome.

In addition, preparing an excuse to leave early or arranging a way to return home in advance can also be examples of action design in A4. These preparations allow a response even if the worst-case outcome occurs.

Actions executed through A4 may produce new information. When this happens, the new information is fed back into A1, and the processes of A2, A3, and A4 may be repeated as needed. For example, if participant information is obtained in response to the question “Who will be there?”, that information becomes a new assumption and may lead to a re-evaluation of priorities and outcome predictions.

The A-Series is not a linear process executed only once. It functions as a cyclical decision-making process that is repeatedly updated in response to changing circumstances and new information.

The purpose of A4 is not to create a perfect plan. Its purpose is to prepare an executable state that can respond to multiple possible outcomes.

Chapter 7: Example

The previous chapters explained A1 through A4 individually. This chapter demonstrates how they function together as a single decision-making process using the same party example.

Situation

You are invited to a party on the weekend and must decide whether to attend.

A1: Assumption Organization

First, the assumptions that form the basis of the decision are organized. The participants are still unknown. The location and time have been confirmed. There is work the next day. There are no health concerns.

At this stage, the factors that may influence the decision are identified. Among them, the fact that the participants are unknown is observed as a factor that may significantly affect later decisions.

A2: Value Criteria

Next, the decision-maker clarifies what should be prioritized. In this example, the value criterion is defined as prioritizing an enjoyable experience.

At this stage, the evaluation criterion for the decision is established. The next question becomes how the participants and atmosphere may affect that criterion.

A3: Branch Prediction

Based on the value criterion established in A2, multiple possible outcomes are maintained. The realistic outcome is the outcome judged most likely based on the participants and past experiences. The best-case outcome is having a highly satisfying experience with compatible people. The worst-case outcome is interpersonal conflict, exhaustion, or damage to relationships.

At this stage, three possible outcomes are maintained simultaneously. The process does not converge on a single outcome.

A4: Action Design

Based on the three outcomes maintained in A3, actions are designed. The objective is to pursue the best-case outcome, avoid the worst-case outcome, and remain capable of responding even if the worst-case outcome occurs.

One resulting action is the question, “Who will be there?” This is not merely a confirmation. It is information gathering designed to increase the likelihood of the best-case outcome while reducing the likelihood of the worst-case outcome. Preparing a way to leave early or arranging transportation in advance may also be part of the action design.

When viewed as a whole, the single question “Who will be there?” can be seen as the result of four distinct processes: assumption organization, value criteria, branch prediction, and action design. If new information is obtained through that question, the information is fed back into A1, and the entire A-Series may be updated accordingly.

The A-Series is a model that decomposes this process into observable stages.

Chapter 8: Connection to the COS Series

The A-Series is a model that decomposes the decision-making process into observable stages. However, the A-Series is not a self-contained system.

In actual cognition, decision-making functions through interactions with other processing systems. Processes such as interpretation, communication, and observation exist alongside

the A-Series or occur before and after it. COS is a framework that integrates these processing systems, and the A-Series constitutes one part of that framework.

In practice, however, some phenomena cannot be fully explained within the observational scope of the A-Series alone.

For example, in dialogue, there are situations where assumption verification appears necessary, yet decisions proceed without that verification being sufficiently performed. This may not be a problem of the A-Series itself. The issue may originate at an earlier stage—how the other person's statement was interpreted before the decision-making process began.

Before any decision is made, there is first a process of interpreting what another person has said. Within COS, this domain is handled by the B-Series (Interpretation System).

The B-Series will be examined in the next paper.